Automated Sperm Assessment Framework and Neural Network Specialized for Sperm Video Recognition

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Highlight

We addressed sperm assessment for reproduction that is an important issue but little-studied in computer vision fields.

- We constructed a video annotated with soft-labels and defined a sperm assessment task.
- We developed an automated framework and sperm-specific network, RoSTFine.
- Experimental results showed the RoSTFine improved on three metrics compared to existing video recognition models.

1. Dataset and Task

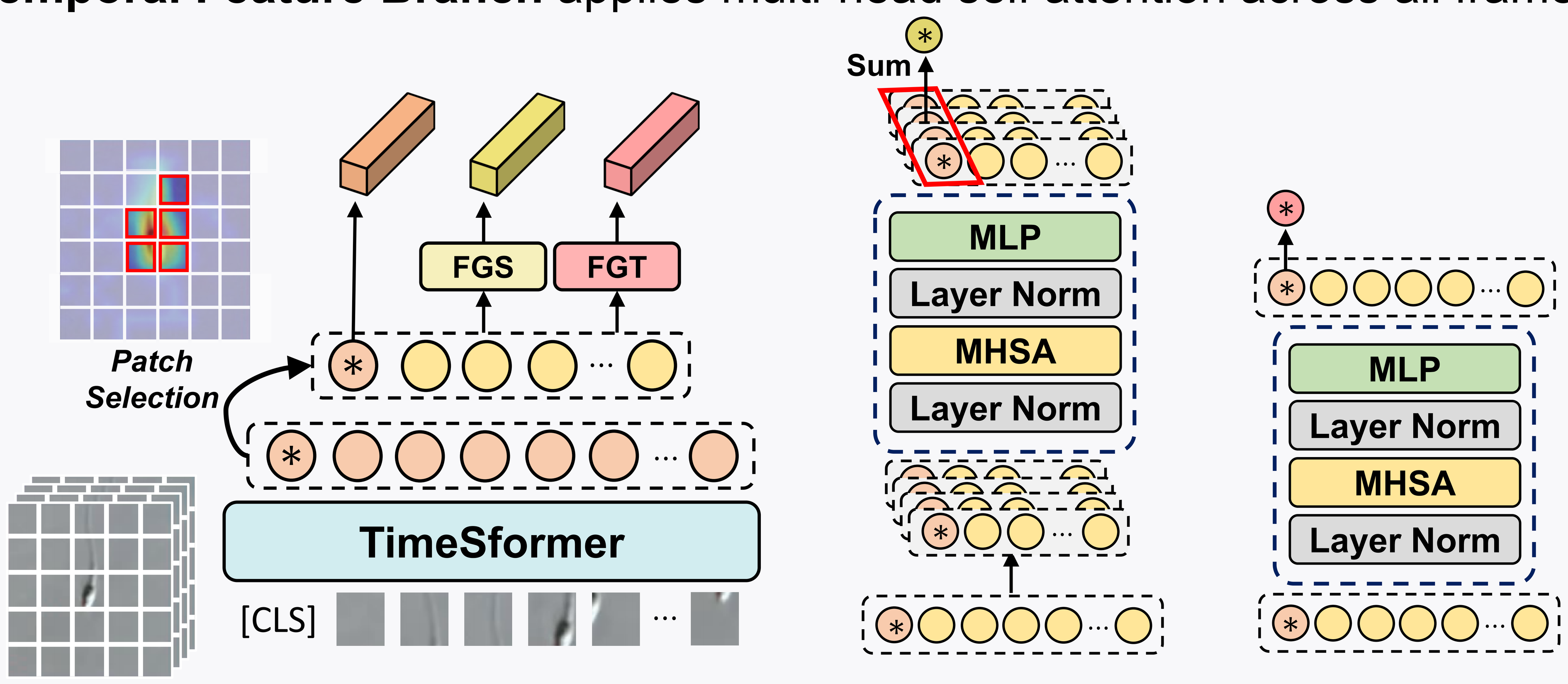
- Our dataset includes 615 videos captured using a microscope.
- Forty experts annotated each sample a grade.
  - Grades are A(Best), B(Good), C (Neither), D (Bad) and E (Worst).
- We refer the label as "*Grade Distribution*" that is a 5-grade histogram.
- We define the sperm assessment task as 5-point regression task.

2. Method

2.1. Architecture

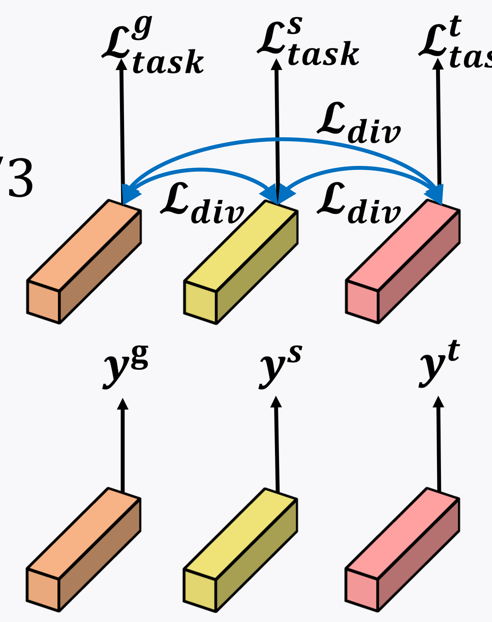
We propose RoSTFine, *Role Separated Transforemr for Fine-grained and Diverse Sperm Feature Extraction.*

- **Patch Selection Module (PSM)** selects k important local patches of each frame based on spatial attention scores.
- **Role-Separated Branch (RSB)** extracts diverse features.
  - **Global Feature Branch** brings the [CLS] patch as a global feature.
  - **Spatial Feature Branch** applies multi-head self attention to each frame.
  - **Temporal Feature Branch** applies multi-head self attention across all frames.



2.2. Training & Inference

- An overall task loss is calculated as the mean of task losses from each feature: $\mathcal{L}_{task} = (\sum_{i\in\{g,s,t\}} \mathcal{L}^i_{task})/3$
- An Orthogonal constraint between two features is calculated as a diversity loss. An overall diversity loss is calculated as the mean of all diversity loss: $\mathcal{L}_{task} = \left(\sum_{i,j\in\{v^g,v^s,v^t\},i\neq j} \boldsymbol{v}^i\cdot\boldsymbol{v}^j/|\boldsymbol{v}^i||\boldsymbol{v}^j|\right)/3$
- The training objective: $\mathcal{L} = \mathcal{L}_{task} + \alpha\mathcal{L}_{div}$
- The final output is the mean of all predictions: $\hat{y} = \left(\widehat{y^g} + \widehat{y^s} + \widehat{y^t}\right)/3$



3. Experiments

Loss Mean Squared Error (MSE) (100×10^{-2}), JS-divergence (100×10^{-2})

Metrics MSE, JS-divergence, Balanced Accuracy (%)

3.1. Main Results (Comparison with Baselines)

- Our RoSTFine$_{\alpha=0,1}$ outperforms the baselines on all metrics.
  ➡ **The RoSTFine architecture (*e.g.*, PSM, RSB) is effective.**

3.2. Visualizing Attention Map

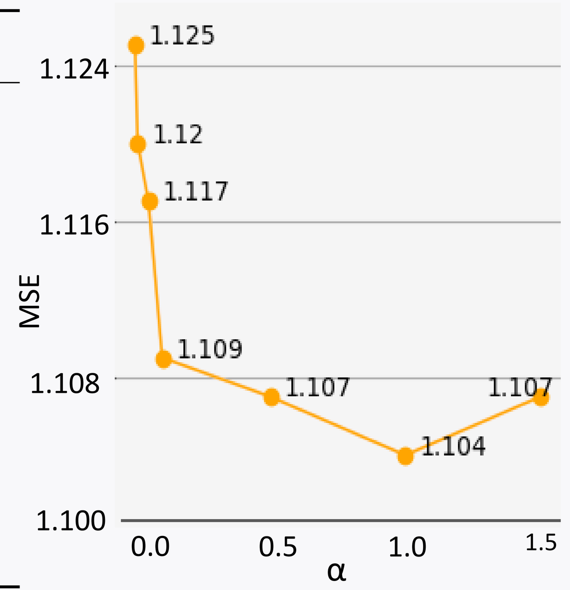
- While TimeSformer attends to a wide area around the sperm, RoSTFine attends strongly to sperm parts (*i.e.*, head and neck) and captures the tail across frames when the tail moves vigorously.

3.3. Ablation Studies

- The performances of only $\boldsymbol{v}^s$ or $\boldsymbol{v}^t$ is higher than that of only $\boldsymbol{v}^g$ (*i.e.*, [CLS] embedding). Moreover, any combinations perform better, and the combination of all features is the best.
  ➡ **The features on FGS/FGT and their combination are effective.**
- The higher α value, the higher the performance in the range of $0 < \alpha < 1$.
  ➡ **The diversity loss is effective to extract diverse features.**
- Our aggregation strategy, separately optimizing each features in training and averaging all outputs in inference , obtains the best performances.

Method	MSE	JS-div.	B.A. Acc.
VGG16†	1.517	5.628	25.44
R3D	1.365	4.978	30.52
R(2+1)D	1.702	7.043	24.30
X3D	1.808	7.186	25.06
I3D	1.376	5.206	30.98
SlowFast	1.346	5.059	30.05
ViViT	1.406	4.987	30.99
TimeSformer	1.186	4.283	33.94
RoSTFine$_{\alpha=0}$	**1.121**	**4.145**	**35.24**
RoSTFine$_{\alpha=1}$	**1.104**	**4.109**	**36.37**

$\boldsymbol{v}^g$	$\boldsymbol{v}^s$	$\boldsymbol{v}^t$	MSE
✓†			1.186
	✓		1.138
		✓	1.145
✓	✓		1.135
✓		✓	1.139
	✓	✓	1.134
✓	✓	✓	1.121



Aggregation Type	MSE	JS-div.
Concat	1.228	4.238
Sum	1.197	4.153
Ours	**1.121**	**4.145**